Rapid Evidence Review

Assessing the benefits and disadvantages of different STI testing schedules for gonococcal, chlamydial and syphilis infections in asymptomatic populations in high-income settings

Michael Traeger, Bernadette Ryan, Benjamin Riley, Joe Givan, Edwina Wright, Vincent Cornelisse

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1 Introduction

1.1 Background and purpose of review

Rates of bacterial STIs (chlamydia, gonorrhoea and syphilis) have risen in many high-income countries, particularly among gay, bisexual and other men who have sex with men (GBMSM) and other key populations. Syphilis is associated with serious complications (neurosyphilis, ocular syphilis, congenital syphilis), while chlamydia and gonorrhoea contribute to pelvic inflammatory disease (PID) and infertility, and may increase HIV acquisition risk.

Guidelines in Australia recommend frequent (3-monthly) asymptomatic STI screening for all GBMSM. However, evidence that such high-frequency screening reduces population-level prevalence or severe outcomes, particularly for chlamydia and gonorrhoea, is limited. At the same time, frequent testing increases antibiotic exposure and may contribute to antimicrobial resistance (AMR).

For other non-GBMSM populations, including women and heterosexual young people, guidelines in Australia (including the RACGP Red Book and Australian STI Management Guidelines) recommend annual screening for chlamydia and gonorrhoea for all young people under 25, and only among those over 25 who are increased risk (e.g., people who have been exposed to an STI, people who have had a recent STI, people travelling to areas of higher prevalence, or has a partner who is a gay or bisexual men, sex worker or who is pregnant).

The purpose of this review is to collate the evidence for whether **current screening versus reduced screening or no screening** should be recommended for gonorrhoea, chlamydia and syphilis infections in the populations of interest.

1.2 Search strategy

Two databases were searched on the 24th October 2025 (Embase and Pubmed) for studies published from 1st Jan 2015 to current, using the following search terms:

- (chlamydia OR gonorrh* OR syphilis OR treponema*); AND
- (screening or testing); AND
- (interval or frequency); AND
- (antimicrobial resistance OR prevalence OR incidence OR sequelae)

Database search results were extracted into Covidence and titles and abstracts were screened by two reviewers for relevance. Citation lists of relevant studies and review articles were also screened for relevant studies published prior to 2015.

1.3 Inclusion criteria and synthesis of findings

We were guided by the GRADE process for searching and synthesising evidence. Given the scope of the evidence being considered, we conducted an evidence review rather than a full systematic review (meant to be comprehensive review of the evidence but not an exhaustive review of all studies meeting a pre-specified criteria).

Studies were included as key studies if they **provided empiric data directly comparing the impact of different screening frequencies for asymptomatic STIs** among a key population. This included randomised controlled trials or observational cohort studies comparing different screening frequencies, as well as ecological studies exploring the association between screening frequency and outcomes. Qualitative studies exploring the values, preferences and acceptability of different screening frequencies were also included as key studies. Key studies were summarised and certainty of evidence was assessed guided by the GRADE framework.

Mathematical modelling studies, and studies which did not directly compare different STI screening frequencies but could provide insights relevant to the topic, hence were included and synthesised narratively as additional evidence where empiric evidence was unavailable.

We did not include studies which reported incidence or prevalence of STIs among key populations without a comparison of these outcomes by different screening frequency. Also, we did not include studies which explored the impact of screening for STIs at a specific anatomic site, or which compared single-site vs multiple-site testing. Screening during pregnancy was not considered within scope for this review.

Given the nature of the studies found from the search, the findings are reported for (1) gay and bisexual men and other men who have sex with men, and trans and gender diverse people (including those with HIV or those using HIV PrEP); (2) the general population (including women, heterosexual men and young people); and (3) Aboriginal or Torres Strait Islander People.

2 Gay, bisexual and other men who have sex with men and transgender people (including those with HIV or using PrEP)

As the current Australian STI Management guidelines recommend frequent (3-monthly) screening for all GBMSM and people using HIV PrEP, the below are summaries of key studies for GBMSM (plus trans and gender diverse people) which report relevant data framed as the desirable and undesirable effects of high-frequency screening. Some studies consider the impact of 3-monthly screening specifically, while others consider the impact of increased screening frequency more generally. Key studies are summarised in Table 1.

2.1 Desirable effects (benefits) of screening

How substantial are the desirable effects of screening? E.g., reduced population-level prevalence, reduced rates of complications, reductions in AMR)

2.1.1 Chlamydia & gonorrhoea

Controlled trials

Vanbaelen (2024)[1] – *Gonoscreen* RCT

In this large RCT of GBMSM and transgender women on PrEP (n=1,104), participants were randomised to 3x3 screening (3-monthly asymptomatic screening in which all positive case were treated) vs no screening (blinded 3-monthly asymptomatic screening where only symptomatic cases were treated). In the screening arm, there were modest reductions in combined CT/NG incidence compared with no screening; however, this was driven by reductions in chlamydia incidence, with no change in gonorrhoea. Given that the methodological bias noted in the non-screening arm (that untreated infections which persisted for more than 3 months were re-counted as new infections), the authors adjusted for repeat infections in sensitivity analyses and found no difference in incidence of either CT or NG between the arms. Overall, the RCT provides little evidence that quarterly screening reduces gonorrhoea incidence, and only weak evidence for chlamydia under some analytic assumptions. However, outcomes were not assessed at the population-level, and moderate levels of bias were present, including likely contamination (sexual encounters) across study arms. Therefore, this study provides little evidence that 3x3 screening leads to population-level declines in prevalence of CT/NG, with equally weak evidence that no screening leads to an increase in prevalence.

Observational studies

Vanbaelen (2025)[2] – Antwerp clinic study

In this single-centre observational study which captured data on 3,955 GBMSM between 2019-2024, authors measured change in CT/NG incidence and antibiotic prescribing during a period where testing recommendations changed from quarterly to no screening for MSM. CT/NG incidence was higher in the early period associated with higher rates of screening, implying that more frequent screening allows earlier detection of STIs. The rate of symptomatic gonorrhoea infections was unchanged between the earlier period of high-frequency screening compared to the period of lower/no screening; implying no evidence that high-frequency screening was associated with fewer symptomatic infections through decreased prevalence. Many biases are presented in this analysis, as screening changes occurred only at a single clinic and authors did not measure impact on population-level prevalence.

Jongen (2023)[3] – *AMPrEP* retrospective analysis

In this retrospective observational cohort study of 366 PrEP users in Amsterdam (AMPrEP), authors estimated the proportion of CT/NG diagnoses which would be delayed if 3-monthly screening was reduced to 6-monthly screening. The authors estimated that 6-monthly screening would delay 52% of asymptomatic CT/NG diagnoses, with that majority (73%) of delayed diagnoses being among GBMSM who reported subsequent condomless sex. This implies that more frequent screening (every 3-months) allows earlier detection, but no data were available on population-level prevalence in this study.

Ecological studies

Marcus et al (2021)[4] - European ecological analysis

This large multi-country survey (EMIS Study) conducted across 46 countries, data were included from 156,018 (2010) and 125,837 (2017) GBMSM. The study found that self-reported diagnoses of CT/NG increased from 2010 to 2017 by 76%. Increases were most pronounced for asymptomatic infections, which increased by 142%. Multivariable modelling showed that much of this rise was explained by greater and more comprehensive screening rather than a reduction in underlying transmission. Higher screening rates were positively associated with higher reported symptomatic gonorrhoea/chlamydia, suggesting that screening did not suppress prevalence and may have increased detection without reducing underlying infection pressure. Overall, this ecological study provides no evidence that increased asymptomatic screening among MSM reduced the prevalence or incidence of chlamydia (or gonorrhoea).

Kenyon et al (2019)[5] – European ecological analysis

This ecological analysis compared country-level intensity of STI screening (self-reported via an internet-based survey of >160,000 MSM conducted in 2010 [EMIS Study]) and incidence and prevalence of CT/NG (taken from surveillance data from 2010 and the literature from 2010-2015) across 23 European countries. The analysis found no evidence that more intensive CT/NG screening among GBMSM at the country level was associated with lower prevalence or incidence of either infection. Higher proportions of GBMSM reporting anal or urethral STI screening were actually correlated with higher national incidence rates of both chlamydia and gonorrhoea. The study therefore provides no indication of desirable population-level effects as screening did not appear to reduce infection burden in GBMSM populations, even in countries with very high screening uptake.

Modelling studies

One modelling analysis (**Buyze et al**)[6] found that increasing NG screening coverage among GBMSM in Belgium would produce only small reductions in gonorrhoea prevalence, even under very high screening levels (e.g. if 50% of the GBMSM population were screened). Screening half the population only lowered pharyngeal prevalence from 11.9% to 10.2%, urethral prevalence from 7.6% to 6.5%, and rectal prevalence from 14.7% to 12.4% after 10 years despite a major increase in screening intensity. Overall, the study indicates that although screening can reduce prevalence slightly at very high coverage levels, the effect on prevalence is

small, and the authors question whether these gains outweigh the substantial increase in antibiotic exposure required to achieve them.

A network-based model (Jenness et al)[7] using data from sexual networks of MSM in Atlanta, USA, explored the impact of HIV PrEP implementation and associated increases in screening on population-level CTNG among GBMSM. In this study, authors estimated that if PrEP coverage reached 40%, with a 40% increase in risk compensation associated with PrEP uptake, 42% of NG and 40% of CT could be prevented over 10 years. Increasing screening from 6-monthly to 3-monthly was project to prevent an additional 50% of infections through increased detection and treatment asymptomatic rectal infections, reducing onward transmission.

2.1.2 Syphilis

No RCTs directly compared the impact of syphilis screening frequency on population-level syphilis outcomes such as incidence and prevalence among GBMSM. In the Gonoscreen RCT (Vanbaelen 2024)[1] and the Antwerp clinic study (Vanbaelen 2025)[2] where CT/NG testing was removed, syphilis screening recommendations were unchanged and remained at 3-monthly screening intervals.

Observational studies

Jongen (2023)[3] – *AMPrEP* retrospective analysis

In this retrospective observational cohort study of 366 PrEP users in Amsterdam (AMPrEP), authors also estimated the proportion of syphilis diagnoses that would be delayed if 3-monthly screening were reduced to 6-monthly screening. The authors estimated that 6-monthly screening would delay about 52% of syphilis diagnoses by three months. This implies that more frequent screening (every 3-months) allows earlier detection, but no data were available on population-level prevalence in this study. Authors also state that "While there is a potential for serious sequelae of syphilis if left untreated, the likelihood of these sequelae within a time frame of 6 months is low."

Chow et al (2017)[8] – *ACCESS* surveillance data

In an observational study of retrospective surveillance data from Australia including over 350,000 clinics visits by GBMSM, authors found that increases in testing (both testing coverage and frequency) from 2007 to 2014 was association with increased detection of asymptomatic infectious syphilis and relative reductions in secondary syphilis among HIV-negative and HIV-positive GBMSM. While the mean number of tests per year only reached between 1.3-1.6 for HIV-negative GBMSM and 1.6-2.3 for HIV-positive GBM (well below quarterly screening rates currently recommended), the study shows that more frequent and widespread syphilis screening led to earlier detection of infections, which in turn reduced the number of cases progressing to more serious, symptomatic, and highly infectious stages such as secondary syphilis. The correlation with reduced secondary syphilis was slightly stronger for increases in testing frequency (rho=-0.93; very strong) compared to increases in testing coverage (rho=-0.87; very strong). This study provides moderate evidence that more frequent screening for syphilis among GBMSM may reduce the rates of serious sequalae such as ocular and neurosyphilis, however the study did not measure this different between 3-monthly vs 6-monthly vs annual screening specifically.

Ecological study

Marcus et al (2021)[4] – European ecological analysis

This large multi-country survey conducted across 46 countries, found that self-reported diagnoses of syphilis increased from 2010 to 2017 by 83%. Increases were most pronounced for asymptomatic infections, which increased by 145%. Screening frequency and increased screening uptake were strongly associated with higher overall syphilis diagnosis rates, which the authors stated reflects more active case-finding. However, unlike CT/NG, higher national screening rates were not associated with increased symptomatic syphilis, suggesting that increased testing primarily uncovered latent or early infections rather than driving reinfection or symptomatic disease. Authors suggest that increased screening contributed to earlier detection of asymptomatic infections.

Modelling studies

Several modelling studies explored the impact of different screening strategies on population-level syphilis.[9-11] Modelling consistently shows that more frequent syphilis screening (typically every 3–6 months) among GBMSM may lead to substantial population-level reductions in infectious syphilis prevalence by shortening the average duration of infectiousness and interrupting transmission chains. Additional modelling evaluating STI screening within HIV-prevention cohorts also found that when screening intervals are shortened, a substantial share of otherwise undetected asymptomatic early syphilis cases are diagnosed earlier, narrowing the window for onward transmission. Taken together, modelling studies suggest that higher-frequency syphilis screening may provide tangible population-level benefits, including fewer infectious cases, earlier treatment, and less progression to late-stage syphilis.

Other considerations

Multiple studies show that syphilis is more likely to be asymptomatic in individuals with prior infection, including GBMSM,[12,13] highlighting the importance of screening to prevent progression to secondary and tertiary syphilis with potentially significantly harmful sequelae.

2.2 Undesirable effects (harms) of screening

How substantial are the undesirable effects of screening? (e.g., increased antibiotic consumption, increased AMR, symptomatic infections)

2.2.1 Chlamydia & gonorrhoea

Randomised trials

Vanbaelen et al. (2024)[1] - *Gonoscreen* RCT

In this RCT of GBMSM and transgender women taking PrEP (n=506) randomised to quarterly screening vs no screening, participants randomised to quarterly screening received a significantly higher rate of antibiotic prescription, largely due to treatment of asymptomatic rectal/pharyngeal infections detected through screening. The rate ratio of antibiotic prescribing in the non-screening arm compared to the screening arm was 0.79 (0.72-0.86) for azithromycin, 0.56 (0.43-0.74) for ceftriaxone and 0.55 (0.52-0.59) for doxycycline. There were no serious adverse events in either arm of this study. This RCT provides high-quality evidence that frequent screening increases antimicrobial exposure at the individual-level without clear incidence benefits for reducing gonorrhoea.

Observational studies

Vanbaelen et al. (2025)[2] – Antwerp clinic study

In this single-centre observational study including 3,955 GBMSM which compared incidence of CT/NG and symptomatic infections over a period of declining screening rates at a large PrEP clinic, authors reported that real-world reductions in screening frequency led to marked decreases in ceftriaxone, doxycycline, and azithromycin prescriptions, while symptomatic STI incidence did not increase during the period of lower screening. Ceftriaxone, doxycycline, and azithromycin prescriptions decreased (rate ratio 0.95, 95% CI 0.91–0.99; p=0.023, 0.47, 0.43–0.52; p<0.0001, and 0.90, 0.87–0.94; p<0.0001, respectively). There was no increase in the incidence of symptomatic N gonorrhoeae (IRR 0.98, 95% CI 0.92–1.03), C trachomatis (0.95, 0.88–1.01), or lymphogranuloma venereum (0.95, 0.84–1.09).This study provides high certainty evidence that reducing screening frequency reduces antibiotic prescribing, and moderate certainty that reducing frequency does not increase asymptomatic infections (although not measured at the population-level).

Jongen et al. (2023)[3] – *AMPrEP* retrospective analysis

Although not measuring antibiotic use directly, authors used retrospective cohort data to estimate that a biannual screening scenario would delay diagnosis in up to half of all CT/NG infections, implying that quarterly screening would substantially increase treatment episodes and antibiotic exposure, as many infections may be detected (and treated) earlier and more often.

Wong et al. (IUSTI 2025; manuscript unpublished) - *ACCESS* surveillance data

This observational study of retrospective surveillance data from 65,261 GBMSM attending sexual health clinic across Australia explored trends in screening and ceftriaxone prescribing. Authors disaggregated outcomes by whether they were screening-related (screening in the absence of symptoms and associated treatment) or non-screening related (testing of people with symptoms or who were contacts of STIs and associated treatment). Overall, total ceftriaxone prescribing in this cohort doubled between 2016–2023, but non-screening-related prescribing (prescriptions given for gonorrhoea cases detected through symptomatic presentation or partner treatment) remained stable, meaning the increase was almost entirely attributable to screening-related diagnoses. The rate of screening-related ceftriaxone prescription (ceftriaxone prescribed following a diagnosis which arose from asymptomatic screening) rose from 0.036 to 0.119 doses/person-year, increasing from 31% to 56% of all ceftriaxone use. At the individual-level, GBMSM who were screened ≥4 times per year had 5.47 times the rate of screening-related ceftriaxone prescriptions compared with those screened annually. This strongly reinforces previous concerns that intensive screening is associated with antimicrobial consumption without clear demonstration of incidence benefit. The authors suggest that reductions in screening could decrease ceftriaxone consumption by up to 56%.

Ecological studies

Kenyon et al. (2019)[5] – European ecological analysis

In this ecological analysis of data from 23 European countries, authors demonstrated higher screening intensity correlates with higher incidence of CT/NG, with authors suggesting this is likely because screening drives increased case detection and antibiotic use without reducing prevalence. Given the ecological nature of this data, many biases are present and certainty is low.

Van Dijk et al (2020)[14] – US ecological analysis

This ecological study compared STI screening intensity among GBMSM across 21 U.S. states, using survey data on self-reported testing rates and gonococcal surveillance data. States with more intensive screening had higher gonococcal MICs for ceftriaxone and cefixime (rho=0.42), indicating reduced susceptibility and a potential acceleration of resistance development in NG isolates This study provides low-certainty evidence that higher STI screening intensity among GBMSM may create selective pressure for resistant strains.

2.2.2 Syphilis

No studies directly reported undesirable effects of frequent screening for syphilis, including any treatment side effect, adverse events or behavioural risk compensation. No study showed an increase in syphilis incidence associated with screening.

While there is no evidence of penicillin resistance in *Treponema Pallidum,* screening for syphilis as part of routine comprehensive STI testing may indirectly increase antibiotic exposure and increase selection pressure for AMR in gonorrhoea, but this was not measured directly in any study.

Across studies, there is no direct evidence that frequent syphilis screening is associated with undesirable effect or harms among GBMSM.

2.3 Certainty of Evidence

What is the overall certainty of the evidence of effects?

Overall, the certainty of evidence for substantial desirable population-level benefits of high-frequency asymptomatic CT/NG screening among GBMSM is low.

The only RCT (Vanbaelen 2024)[1] showed a small reductions in chlamydia (and none for gonorrhoea), however in sensitivity analyses there was no difference between arms for either infection, with important risks of bias, contamination between arms, and no direct population-level outcomes. Observational and ecological studies are consistently non-supportive of reductions in incidence or prevalence, and modelling by Buyze 2018 suggests only modest prevalence reductions even in a setting with unrealistically high coverage, while modelling by Jennes et al suggests some reductions in CT/NG may be possible with high PrEP/screening coverage (although model was done prior to large scale PrEP rollout). Across studies there are moderate-high risks of bias, indirectness, and inconsistency.

For syphilis, one large time-trend study (Chow 2017) and multiple modelling studies suggest that more frequent screening is plausibly associated with earlier detection and fewer cases of stages associated with complications (e.g. secondary syphilis), but data are observational or modelled, without direct comparisons of specific screening intervals. Therefore, there is low-moderate certainty for clinically important reductions in complications.

In contrast, the evidence that high-frequency CT/NG screening substantially increases antibiotic consumption is of moderate to high certainty (high-quality RCT plus large observational datasets showing large, consistent increases in ceftriaxone and other antibiotics when screening is intensified). The association between screening intensity and rates of AMR is also supported by ecological studies.

2.4 Values, preferences & acceptability

Berners-Lee et al. (2025)[15] – Qualitative study of GBMSM in England

Qualitative interviews with 22 GBMSM in England show strong attachment to current high-frequency screening (3-monthly). Many participants express discomfort and "unease" about reducing screening frequencies, despite awareness of AMR. Trust, habit, perceived safety, and alignment with self-identity as responsible health-seeking individuals underpin preferences for frequent testing. Reductions were more acceptable at 6-monthly, but not at lower frequencies.

Wardley et al. (2023)[16] – Qualitative study of Australian GBMSM

In a qualitative study involving semi-structured interviews with 12 GBMSM, participants were hesitant about reducing the screening frequency and reported that testing gave them a sense of security. While MSM recognised antimicrobial resistance was a concern, it did not impact their sexual behaviour, with many participants stating they would rather continue to take antibiotics to treat infections rather than adopt preventative measures such as condom use. Positive attitudes towards screening interval changes are more likely when PrEP patients are informed about the risks and benefits of sexual healthcare recommendations

Ludwick et al. (2025)[17] – Qualitative study of Australian clinicians

Qualitative interviews with 16 clinicians (12 physicians (including 8 GPs) and 4 nurse practitioners) showed cautious openness to reducing CT/NG screening frequency but emphasised needing clear evidence that reduced screening would not increase transmission, especially to women. They anticipated community resistance, noting GBMSM often view frequent screening as routine, reassuring, and part of sexual health identity. Clinicians also described a strong professional obligation to treat any positive result, with many uncomfortable about non-treatment due to fears of onward transmission and patient distress. Although aware of AMR concerns, most prioritised preventing transmission. Clinicians supported incremental change to reducing screening frequency if recommendations were to be reduced, paired with community engagement and clear communication.

Ludwick et al. (2025)[18] – Global crowdsourcing study

This global crowdsourcing study of 99 STI experts did not find consensus on whether screening for asymptomatic CT/NG in GBMSM should be reduced, but it showed moderate support for shifting from 3-monthly to 6-monthly screening, driven largely by concerns about resource burden, including clinic workload and costs (especially within PrEP programs). Attitudes were divided on core clinical questions: whether asymptomatic CT/NG infections cause meaningful harm, whether they require treatment, and whether frequent screening actually reduces population-level prevalence. Many participants expressed uncertainty or unfamiliarity with emerging evidence showing limited population-level impact of frequent CT/NG screening, and participants had contrasting views on AMR. Despite disagreements, there was strong agreement that reducing CT/NG screening must not compromise frequent HIV and syphilis testing, and that any guideline change would require careful community engagement to avoid confusion and preserve trust, given decades of messaging encouraging regular STI screening. Concerns were raised about unintended consequences, such as impacts on other STIs, potential misperceptions that changes are cost-cutting, and stigma risks. Overall, resource efficiency was the area of greatest agreement, suggesting that any movement toward less frequent screening would likely need to be incremental and accompanied by clear messaging and strong safeguards for HIV and syphilis detection.

Vanbaelen et al. (2024)[1] – Qualitative substudy of Gonoscreen RCT

In a qualitative sub-study of the Gonoscreen RCT, three focus group discussions were held, each with three to five randomly selected participants, also included were two in-depth interviews with PrEP users who were invited to the main study, but declined. Participants expressed mixed attitudes toward the idea of non-screening for asymptomatic CT/NG. Some viewed these infections as mild, self-limiting, and unlikely to cause individual harm, making routine testing seem unnecessary. For these participants, the absence of symptoms and perceived low clinical severity reduced the value they placed on screening. Others, however, identified ongoing transmission to sexual partners as a major concern, describing feelings of guilt, risk, and irresponsibility when imagining not being tested or treated. Preferences were also shaped by sexual behaviour, with individuals with higher partner numbers or low condom use preferring frequent testing, whereas those with fewer partners or in more stable relationships felt comfortable with less frequent screening. Perceptions of antimicrobial resistance varied widely. Some participants considered AMR a meaningful concern and wished to limit unnecessary antibiotic use, while others reported limited knowledge of AMR or had difficulty weighing the risk of AMR against the risk of transmitting an asymptomatic infection. Several participants noted that their attitudes shifted over the course of the discussion as they learned more about the natural history of CT/NG and AMR, sometimes becoming more open to reduced screening. Overall, acceptability of non-screening depended on risk perception, sexual practices, sense of responsibility to partners, and understanding of AMR, suggesting that preferences are heterogeneous and influenced by knowledge and context.

Stahlman et al (2015)[19] – Qualitative study on acceptability of syphilis interventions

Among 19 GBMSM from the US who had repeated syphilis infections and underwent in-depth interviews, frequent syphilis screening (every 3–6 months) was generally viewed as acceptable, expected, and aligned with participants' perceptions of their ongoing risk. Most respondents were already testing every 3–6 months, and many expressed willingness to test even more frequently if testing were made easier or more convenient. Automated reminders (text or email) to test every 3 months were acceptable to over half of participants, with strong preferences for low-burden, discreet communication methods. Participants found home-testing kits or web-based systems acceptable.

2.5 Summary table of key studies on effect of screening frequency in men who have sex with men

	Author (year)	Population / Setting	Intervention / Comparison	Study Design	Key Findings	Notes on Certainty of Evidence
RCT	Vanbaelen et al. (2024) *Gonoscreen study*	MSM & transgender women on PrEP in Belgium (5 HIV centres), N=1,014	3×3 screening (every 3 months, 3 sites) vs non-screening (results masked unless symptomatic)	Non-blinded randomised controlled trial	Non-screening **arm** had higher combined NG/CT incidence (IRR 1.32), but no difference in gonorrhoea incidence. Sensitivity analysis accounting for untreated infections showed no difference in incidence in either organism. Screening arm had significantly higher antibiotic use.	High certainty (RCT) for screening increases antibiotic consumption. Moderate certainty that screening does not reduce NG incidence and has no effect on CT/NG incidence due to sensitivity analyses showing null findings when untreated cases were accounted for. High internal validity; direct population; precise estimates.
Observational cohort	Vanbaelen et al. 2025 [2]	MSM using HIV PrEP, N=3,955 Single centre in Antwerp, Belgium. 2019–2024	Routine CT/NG screening for MSM on PrEP was discontinued in early 2023. Comparator: Earlier period with routine (systematic) screening.	Retrospective cohort using routinely collected clinical data; Poisson models estimating IRRs for infections and antibiotic prescriptions.	CT incidence decreased from 198.2 to 60.7 per 1,000py, yearly IRR 0.84 (95% CI 0.81–0.87), p<0.0001. Symptomatic infections were unchanged: NG IRR 0.98 (95% CI 0.92–1.03); CT IRR 0.95 (0.88–1.01); LGV IRR 0.95 (0.84–1.09). Antibiotic prescriptions decreased: ceftriaxone RR 0.95 (0.91–0.99), p=0.023; doxycycline RR 0.47 (0.43–0.52), p<0.0001; azithromycin RR 0.90 (0.87–0.94), p<0.0001. Sensitivity analyses broadly consistent.	High certainty for declines in screening lead to substantial reduction in antibiotic use. Low-moderate certainty for screening reduction does not lead to more symptomatic cases (potential confounding; single-clinic; secular trends may influence findings). Risk of bias from retrospective, single-centre design, but large sample with sensitivity analyses performed.
	Jongen et al (2023)[3]	MSM on PrEP in Amsterdam (AmPrEP study) N=366	Modelled estimate of impact of reduction in screening from 3-monthly to 6-monthly	Retrospective observational cohort study	Among 366 participants with 1,183 STIs of which 932 were asymptomatic, 483 (52%) asymptomatic diagnoses would have been delayed at 364 study visits. 73% of visits where an asymptomatic STI would have been missed with reduced screening were followed by periods of condomless sex with casual partners.	Moderate certainty for reducing screening would delay some diagnoses. However outcome is “diagnoses delayed” rather than population-level transmission or incidence (indirectness). Observational design with risk of bias.
	Tang et al. (2015)[20]	MSM on PrEP in the US PrEP Demo Study N=557	Modelled estimate of impact of reduction in screening from 3-monthly to 6-monthly	Retrospective observational cohort study	If 3-monthly screening had been reduced to 6-monthly screening, 34.3% of gonorrhoea, 40.0% of chlamydia and 20.4% of syphilis diagnoses would have been delayed. 89.2% of asymptomatic cases reported condomless anal sex.	Moderate certainty for reducing screening would delay some diagnoses. Findings consistent with other cohorts. Limitations include modelling component and indirect outcome (delay in diagnosis, not population-level incidence).
	Wong et al. (unpublished)	MSM in Australia attending a network of sentinel clinic N=65,261	Analysed cohort-level rates of screening-related (no symptoms or STI contact) and non-screening related (done due to symptoms or being an STI contact) testing and antibiotic consumption from 2016-2023.	Retrospective observational cohort study Sentinel surveillance data	Total ceftriaxone prescribing doubled between 2016–2023, but non-screening-related prescribing (prescriptions given for gonorrhoea cases detected through symptomatic presentation or partner treatment) remained stable, meaning the increase was almost entirely attributable to screening-related diagnoses. Screening-related ceftriaxone rose from 0.036 to 0.119 doses/person-year, increasing from 31% to 56% of all ceftriaxone use.	Moderate certainty for increased screening increases antibiotic use. Causal inference limited by ecological and observational design, but effect direction (screening strongly associated with increased ceftriaxone use) is consistent and plausible. Very large dataset strengthens precision. High certainty for describing trends; low certainty for causality.

	Author (year)	Population / Setting	Intervention / Comparison	Study Design	Key Findings	Notes on Certainty of Evidence
Ecological	Kenyon et al (2019)[5]	MSM in 23 European countries; data from EMIS & ECDC	Countries with higher screening intensity vs lower screening intensity	Ecological correlation study	No negative association between screening intensity and prevalence of gonorrhoea or chlamydia; positive association with national incidence (rho 0.74–0.78). Suggests screening intensity does not reduce prevalence nor incidence.	Low/very low certainty for causal effects due to ecological design, risk of confounding, and indirectness. Findings provide supportive contextual evidence but cannot infer whether screening causes higher incidence or vice versa.
	Van Dijk et al (2020)[14]	MSM in 21 US States	Compared STI screening intensity among MSM survey respondents with gonococcal susceptibility to azithromycin, ceftriaxone and cefixime in 21 US states	Ecological correlation study	A positive ecological association was found between STI screening intensity and geometric mean gonococcal minimum inhibitory concentration for ceftriaxone (rho=0.42, p=0.031) and cefixime (rho=0.42, p=0.029), but not for azithromycin (rho=0.31, p=0.11).	Low/very low certainty. Ecological correlations between screening intensity and AMR trends are hypothesis-generating only; multiple uncontrolled confounders likely.
	Marcus et al (2021)[4]	MSM in Europe	Compared self-reported STI diagnosis among survey respondents in 2010 and 2017, with country-level screening rates	Ecological correlation study	Between 2010 and 2017, overall diagnoses with gonorrhoea/chlamydia and syphilis increased by 76% and 83% across countries. Increases were more pronounced for asymptomatic compared to symptomatic infections. Gonorrhoea/chlamydia increases were mainly explained by more screening and a change in testing performance.	Low/very low certainty. Ecological comparisons with strong confounding. Self-reported STI diagnosis.
Qualitative	Wardley et al. (2023)[16]	MSM on PrEP in Australia, qualitative interviews (N=13)	No comparison. Interviewed attitudes related to reducing screening frequency	Qualitative thematic study	MSM were initially hesitant to reduce screening, as they felt testing provides peace of mind, relates to health and community protection. Participants acknowledged AMR concerns, but concerns did not influence behaviour. Reductions in screening may be acceptable with transparent communication and trusted clinicians.	High certainty (qualitative evidence) for concerns around reducing screening frequency in MSM. Clear themes, acceptability of reduced testing explored, directly relevant population (Australian MSM).
	Berners-Lee et al. (2025)[15]	22 MSM and 8 professional stakeholders in the UK	No comparison Qualitative interviews	Qualitative thematic study	Among MSM responses to discontinuing asymptomatic NG and CT screening tended to be negative, while professional stakeholder opinions were mixed. Reducing screening frequency to 6-monthly was more acceptable to both groups.	High certainty for qualitative insights (MSM population plus diverse stakeholder perspectives).
	Ludwick et al. (2025)[17]	Australian clinicians working in sexual health and general practice (n=12)	Consideration of reducing CT/NG screening frequency for GBMSM	Qualitative interview study	Clinicians were cautiously open to less frequent CT/NG screening but required clear evidence that reduced screening would not increase transmission. Support existed for incremental change only, with clear communication and community engagement.	High certainty for qualitative evidence. Insights on Australian clinician attitudes, perceived risks, and implementation barriers. Applicable to acceptability and feasibility considerations.
	Ludwick et al. (2025)[18]	99 global STI experts (clinical, public health, and research professionals)	Attitudes toward reducing CT/NG screening among GBMSM, including shifting from 3-monthly to 6-monthly intervals	Global crowdsourcing study	No expert consensus on reducing screening was achieved, although there was moderate support for shifting from 3-monthly to 6-monthly screening due largely to concerns about resource burden in clinics. Strong agreement that HIV and syphilis testing frequency must not be reduced, and that careful comminute engagement would be needed to avoid confusion after years of messaging around frequent STI testing.	Moderate certainty. Large expert sample provides breadth, but crowdsourcing may introduce sampling and response biases.

3 General population (including women, heterosexual men and young people)

3.1 Desirable Effects of Screening

3.1.1 Chlamydia and gonorrhoea

Controlled trials

Two large cluster RCTs have evaluated the effect of screening for CT in general populations (Hocking et al 2018[21] & van den Broek et al 2012[22]) and both found no significant impact of screening on CT prevalence. One immediate vs delayed RCT (Oakeshott et al 2010)[23] explored the impact of CT screening on pelvic inflammatory disease (PID) at 12 months after screening. No RCTs have been conducted to evaluate the efficacy of screening for NG prevalence in general populations.

Hocking et al. (2018) [21] - ACCEPt RCT

In this cluster-randomised trial, 130 clinics within 52 rural Australia towns (clusters) were randomised to receive a multi-faceted, clinic-based chlamydia testing intervention or usual care. The intervention package encouraged clinicians to offer annual chlamydia testing to eligible patients. Over a mean of 3.1 years, 93,828 young adults attended intervention clinics and 86,527 attended control clinics. In this RCT, there was no difference in the reduction in CT prevalence among clinic-attendees over time between the two cluster arms (5.0% to 3.4% in intervention arm and 4.6% to 3.4% in control arm; OR=0.9 [0.5-1.5]). However, increasing opportunistic chlamydia testing in primary care led to a modest but meaningful reduction in hospital-diagnosed PID, with cases occurring 40% less often in intervention clusters than in controls (absolute different -13.7 per 10,000 women; risk ratio 0.6 [0.4–1.0], p=0.044). Although clinic-diagnosed PID did not differ between groups, the lower rate of hospital-treated PID suggests that expanded annual testing, through earlier detection and treatment of asymptomatic infections, may help prevent progression to more severe PID requiring inpatient care. Overall, the study supports a desirable effect of increased screening in reducing severe PID, though the absolute reduction was relatively small.

van den Broek et al. (2012)[22] – Chlamydia Screening Implementation Programme

In this step-wedge implementation RCT, the Chlamydia Screening Implementation Programme was implemented across three regions of the Netherlands from March 2008 to February 2011. Over 300,000 women and men aged 16-29 years listed on registries were invited to use an internet site to request a self-collection STI kit for chlamydia. This large, register-based screening trial found little evidence of population-level benefit from annual chlamydia screening at the participation levels achieved (participation rate was 16%), but it did show some desirable effects among individuals who repeatedly engaged in screening. While overall chlamydia positivity in intervention areas remained essentially unchanged compared with controls (4.3% at baseline vs 4.1% after three rounds of invitations), the small subgroup who participated in all three yearly rounds experienced a substantial decline in positivity from 5.9% to 2.9%, suggesting that regular testing and treatment can meaningfully reduce individual infection risk when uptake is consistent. Estimated prevalence also declined modestly in some regions, indicating that screening may confer localised benefits where participation is higher or more targeted. The authors conclude that “participation was too low to reduce chlamydia transmission in the target population over three years of the intervention”. The impact on self-reported PID could not be assessed reliably as response rates to questions on PID were low.

Oakeshott et al. (2010) [23] - POPI Trial

In the POPI trial of 2,529 sexually active female students aged 16-27 in London, participants provided a vaginal swab sample at baseline, with samples randomly allocated to immediate CT testing and treatment or storage and delayed testing after one year. Baseline prevalence of CT was 5.4%. Among women who were chlamydia-positive at baseline, incidence of PID at 12 months was 9.5% in untreated control arm vs 1.6% in the immediate test and treat arm, suggesting a significant reduction in PID after 12 months among those who received CT screening and treatment. However, the majority of PID cases (79%, 30/38) at 12 months

occurred in women who were negative for chlamydia at baseline, suggesting that the absolute benefit of screening for CT annual to reduce PID may be small.

Observational studies

Dunville et al. (2018)[24]

In this school-based program in Detroit, voluntary STI screening was offered across 4 high schools, with a high student participant rate (5,423 of 6,836 students provided a sample over 5 years, about 80% participation across schools). Annual chlamydia screening in high school students was associated with a sustained and substantial reduction in chlamydia prevalence over five years, falling from 10.24% to 6.27% overall (39% relative decline) with greater decreases among female students (from 16.0% to 9.8%). Each participating school experienced declines, and the reductions were statistically significant overall and among females. The findings suggest that when delivered in a high-prevalence setting with strong participation and high treatment rates, repeat population-level screening may meaningfully reduce the prevalence of chlamydia, thereby lowering the risk of downstream complications such as PID (although PID was not measured in this study).

Owusu-Edusei et al. (2010)[25]

In this study of a large U.S. insurance claim cohort with over 207,000 women, authors performed an exploratory time-series analysis from 2001 to 2006. Increases in population-level chlamydia screening were consistently associated with subsequent reductions in PID, suggesting that more frequent or broader screening can help prevent rates of PID at the population level. Over the six-year period, screening rates more than doubled while PID diagnoses fell by about half, and time-series modelling showed that a 10% increase in the growth rate of screening was followed by a 3–4% reduction in the growth rate of PID about 4 months later. Although the ecological design limits certainty, the temporal pattern and lagged associations support the desirable effect that expanding chlamydia screening coverage may reduce the population burden of PID through earlier detection and treatment of infections before they progress.

Werner et al. (2025)[26]

This retrospective cohort study of women and heterosexual men attending sexual health clinics in the Netherlands (241,000 consultations among 124,000 individuals), aimed to quantify the potential impact of the new guidelines on removing CT screening for this population in the Netherlands, and only testing symptomatic patients or contacts of STI. By assessing which consultations a test would not have been performed under the new guidelines, authors estimated that removing asymptomatic screening would result in approximately 70% fewer CT tests and 55% fewer CT diagnoses, and therefor a similar reduction in antibiotic prescriptions. Most of these detections occurred in young women under 25, the group with the highest underlying burden. These data imply that CT screening substantially increases case detection, enabling timely treatment and interruption of transmission. Authors suggest that screening may therefore help reduce the duration of untreated infection and may prevent onward spread and potential complications, especially in younger women. However, data are observational and the analysis is exploratory.

Modelling studies

Lewis et al. (2018)[27]

This modelling study was based upon the rollout of England's National Chlamydia Screening Program (2003-2008). The authors used a model-based analysis to estimate changes in prevalence and average duration of infection before, during and after the program. The study found that as testing expanded between 2008 and 2010, estimated prevalence in 15-24 year-olds declined by about 0.66 (0.25-1.37) percentage points, and mean duration of infection was reduced by about 74 days in men and 30 days in women. Prevalence estimates increased once testing rates fell after the program stopped in 2010. This study provides evidence that large-scale population testing may be associated with a reduction in chlamydia prevalence, however other outcomes such as PID were not considered.

Other studies

Although published outside of our literature search inclusion period (2010-2025), a systematic review published in 2009 (Low et al.)[28] that included five randomised trials, one non-randomised comparative study and on time trend study, found that 2 randomised trials demonstrated that register-based screening of women at high risk of chlamydia, and female and male high school students, modestly reduced the incidence of PID in women at 1 year, however the authors acknowledge that 'methodological inadequacies could have overestimated the observed benefits. Overall, the review found low to moderate quality evidence that screening CT was associated with a reduced incidence of PID.

The US Preventive Services Task Force found no studies that directly evaluated the effectiveness of screening for gonorrhea in its current or previous reviews of the benefits and harms of screening.[29]

3.1.2 Syphilis

No studies included in our search directly measured benefits of syphilis screening frequency compared to no screening in heterosexuals.

Other studies

The US Preventive Service Task Force conducted a systematic review in 2022 on the benefits and harms of syphilis screening in heterosexuals[30] and concluded "No studies reported the effectiveness of screening on acquisition or transmission of other STIs or other complications such as tertiary syphilis or neurosyphilis. No studies directly addressed effective screening intervals in the included populations."

3.2 Undesirable Effects of Screening

3.2.1 Chlamydia and gonorrhoea

The large observational and ecological studies show that screening programs among young heterosexuals often detect and treat large numbers of infections, also without strong evidence of significant or sustained population-level declines in prevalence.[22,27] These studies suggest high screening in this population would require high resources and increased antibiotic prescribing.

No key studies directly measured the association between screening frequency and AMR in gonorrhoea among heterosexuals. While also not directly measured in included studies, potential undesirable effects of CTNG screening may include as anxiety or stress related to frequent testing, stigma associated with screening interventions, and the potential for risk compensation, although no evidence for these was found in included studies.

Oakeshott et al.[23]

In this immediate vs delayed treatment study, the low proportion of PID cases that were among participants who were chlamydia-positive at baseline implies that a single screen would prevent a relatively low proportion (~21%) of PID cases at 12 months. Authors suggest that testing and treatment (with associated costs, antibiotic exposure, and potential psychosocial impacts) may prevent relatively few PID events, suggesting that the true progression risk from chlamydia to PID may be lower than often assumed, which may mean the cost-effectiveness and overall value of routine screening may have previously been overstated.

Other studies

Kenyon et al (2020)[31] conducted a historical review of the effect of a mass gonorrhoea treatment campaign in Greenland from 1965 to 1968, by exploring the literature on the incidence of gonorrhoea and AMR over

the same period. The review offers limited evidence of desirable effects from very intensive gonorrhoea screening and treatment. Overall, the review concludes that while intensive, repeated screening and treatment may have produced short-lived reductions in incidence, these effects were modest and unsustained. The campaign was however associated with an increase in the proportion of NG that were not susceptible to penicillin.

We did not specifically search for studies reporting psychosocial harms associated with STI screening. A systematic review[32] conducted in 2010 which included 41 studies, authors reported that across studies there was some evidence that undergoing screening, or having a diagnosis of CT, may cause a small-to-moderate number of people to experience some degree of harm, mainly due to feelings of stigmatization and anxiety about future infertility risk.

The US Preventive Service Task Force conducted a systematic review[29] in 2021 on the benefits and harms of chlamydia screening. The USPSTF reviewed several studies, including 4 recent studies (n = 5666), assessing harms of site-specific chlamydia and gonorrhea testing as well as harms of collection methods in women. The false- positive, false-negative, false alarm, and false reassurance rates varied by anatomical site but were overall generally low across all NAATs and specimen types. The review found no studies which evaluated psychosocial harms related to screening or evaluated effect of screening on changes in risk behaviour.

3.2.2 Syphilis

No studies included in our search directly measured harms of syphilis screening in heterosexuals.

The US Preventive Service Task Force conducted a systematic review in 2022 on the benefits and harms of syphilis screening in non-pregnant[33] and included one study on the emotional stress associated with rapid POC STI testing, including syphilis POC testing:

1 fair-quality, pre-post design study (n = 1097) examining emotional stress associated with rapid POC STI testing. Participants considered to be in high-risk groups completed a questionnaire assessing emotional stress prior to and after testing for HIV, hepatitis C, and syphilis. Factors associated with increased stress included history of injection drug use, Black race, less than a high school education, and single marital status. The study did not compare changes in the levels of emotional stress pretesting vs posttesting.[34]

3.3 Certainty of Evidence

The certainty of evidence for desirable effects of CT screening, including reductions in prevalence, in heterosexual populations is low overall. Of the three RCTs[21-23] included, none provides evidence for a reduction in population-level chlamydia prevalence associated with screening. However, studies show a benefit of a small absolute reduction in PID in some settings. In the ACCEPt trial, there were fewer hospital-diagnosed cases PID in the screening intervention clusters. In the POPI trial, there was a modest reduction PID risk among women already CT-positive, but few total cases prevented and many PID cases arising from incident infections).

The large observational and modelling studies show some evidence that high-coverage screening may reduce prevalence and shorten duration of infection, but these are limited by ecological/observational design. Bias is present across studies (possible contamination across clusters, loss to follow-up, testing outside of study, and school or insurance-based cohorts may not be representative of all heterosexuals) and inconsistency across studies (some settings show declines, others do not). Certainty is low for a modest reduction in chlamydia prevalence in high-coverage, high-burden settings, and low for a small reduction in PID, with very low certainty of any benefit for gonorrhoea.

For undesirable effects, while antibiotic use was not measured directly, the certainty is higher that screening increases antibiotic use through increased detection of cases, and lower for other outcomes such as increased AMR consequences. Increases in chlamydia screening were associated with increased diagnosis and treatment of mostly asymptomatic infections. This provides modest certainty that screening in heterosexual populations increases antibiotic prescribing and resource use.

Evidence that this increased antibiotic exposure leads to clinically important AMR in NG or other organisms in heterosexual populations is indirect and low certainty, relying mainly on biological plausibility and extrapolation from other settings. Psychosocial harms (anxiety, stigma, possible risk compensation) are frequently mentioned but rarely measured, so certainty for these outcomes is very low.

3.4 Summary table of studies on screening in heterosexual people

	Author (year)	Population / Setting	Intervention / Comparison	Study Design / Quality	Key Findings / Data Extracted	Notes on Certainty of Evidence
RCT	Hocking et al. (2018)[21]	Rural Australian towns; men & women aged 16–29 attending primary care clinics. (n=52 clusters; 130 clinics; 180,355 young adults)	Clusters were randomised to receive intervention to increase opportunistic annual chlamydia testing vs SOC	Cluster randomised controlled trial (52 clusters, 130 clinics). Pragmatic real-world design.	Screening intervention increased testing rates (20.1% vs 8.2%) but did not reduce chlamydia prevalence (intervention 5.0% to 3.4%, control 4.6% to 3.4%; OR 0.9, 95% CI 0.5–1.5). Reduction in hospital-diagnosed PID (24.2 vs 37.9/10,000 women; RR=0.6 [0.4-1.0]) but no change in clinic-diagnosed PID or epididymitis.	Moderate–high certainty that increased opportunistic testing coverage does not reduce population-level CT prevalence (high-quality cluster RCT). Low–moderate certainty for a small reduction in hospital-diagnosed) PID
	Van den Broek et al (2012)[22]	The Netherlands (3 regions, Amsterdam, Rotterdam, South Limburg); men and women aged 16-29 years. (n=317,304)	From March 2008 – February 2011, annual chlamydia self-sample mail in kits mailed out to all women and men aged 16-29 on population register.	Controlled trial with randomised stepped wedge implementation in three blocks (82 clusters – Amsterdam, 62 clusters – Rotterdam, 46 clusters – South Limburg)	CT mail-in screening uptake decreased after each invitation (16.1%, 10.8% 9.5%) vs control (13.0%) Chlamydia positivity at 1st intervention block was the same as control (4.3%), 0.2% lower by 3rd block. No significant decrease in positivity was seen after screening rounds in any region or sociodemographic regions. Positivity only fell (5.9%-2.9%) among those who took part in all 3 yearly arms (2.8% of population). PID incidence unchanged.	Moderate certainty that low-uptake register-based screening does not reduce population-level CT prevalence over 3 years. Large RCT but constrained by low participation and possible contamination (testing outside the program). Low certainty for any effect on PID due to sparse, self-reported data and low event numbers.
	Oakeshott et al (2010)[23]	Young (16-27) sexually active female students in London (n=2,529)	Participants provided a vaginal swab sample at baseline and randomly allocated to testing and treatment immediately or after 12 months.	Immediate vs delayed randomised controlled trial. Control arm did not receive baseline test result and were not treated if CT-positive at baseline.	Among women who we positive for chlamydia at baseline, incidence of PID at 12 months was 9.5% in delayed arm (untreated) vs 1.6% in the immediate test and treat arm, highlighting reduction in PID at 12 months. However, 79% of PID cases at 12 months were not CT positive at baseline, indicating small absolute benefit of screening.	Moderate certainty that screening and treating prevalent CT reduces short-term PID risk in women with CT (good trial design, clear effect in subgroup). Low certainty that single-round screening meaningfully reduces overall PID incidence at the population level, because most PID cases arose from new infections and the absolute number prevented was small.
Observational	Lewis et al (2018)[27]	2000-2015; Women and men aged 15-24 years England: Public Health England data, published surveillance data	Large scale opportunistic CT screening was implemented from 2003-2008. Authors modelled CT prevalence and infection duration before, during and after screening program	Modelling, observational, ecological	Large scale CT screening was associated with reduction in CT prevalence (similar rates between men and women) and reduction in average infection duration (duration generally longer in men). However, after the program stopped in 2010, prevalences estimates increased once testing rates fell.	Low certainty overall (modelling + ecological design, unmeasured confounding or possible temporal trend). Moderate certainty that high-volume testing can reduce average duration of untreated CT infection, but low certainty that this translates into substantial, sustained population-level prevalence reduction.

	Dunville et al. (2018)[24]	High school students aged 14-19 years – enrolled in 4 Detroit public schools. (n=6,836)	Voluntary school based annual CT screening.	Observational. CT prevalence was measured over time, from 2020-2011 to 2014-2015.	Uptake of CT screening was high (5,423/6,836; 80%). Annual screening was associated with a 39% decline in CT prevalence, with declines across all schools, and greater declines in females.	Moderate certainty that in high-prevalence, high-participation school setting, repeated screening reduced CT prevalence (large effect, consistent across schools). Internal validity is reasonable but indirectness and potential temporal confounding limit generalisability beyond similar settings. No direct evidence on PID or long-term sequelae.
	Werner et al (2025)[26]	Men and women aged 16+ attending publicly funded sexual health services, The Netherlands; n=124,527	Retrospectively estimated the proportion of CT diagnoses which would have been missed under new guidelines of no asymptomatic screening.	Observational; retrospective analysis of surveillance data	No asymptomatic CT screening would result in; 69.5% less CT tests (60.2% women <25 years), 54.9% positive CT cases undetected and untreated (68.0% women <25 years) Removal of asymptomatic CT screening would lead to substantial reduction in tests, CT detection and antimicrobial consumption.	Moderate certainty that asymptomatic screening substantially increases CT case detection especially in young women (large real-world dataset, direct estimation of missed tests/cases). Low certainty about downstream impacts on transmission or complications.
Ecological	Owusu-Edusei et al. (2010)[25]	Cohort of 207,695 privately insured women aged 15–39 years in the U.S. Followed from 2001–2006 using MarketScan claims data	Compared population-level CT/NG screening intensity over time with rates of PID 4 months later. Exposure = monthly chlamydia screening rate; outcome = monthly PID diagnosis rate4 months later.	Ecological, exploratory time-series analysis. Outcome (PID) based on ICD-9 codes; limited ability to adjust for individual-level confounding; only privately insured population	In time-series regression, a 10% increase in the growth rate of chlamydia screening 4 months earlier was associated with a 3.6% decrease in the growth rate of PID diagnoses (β–0.36; p<0.05), consistent with screening contributing to reduced PID at population level.	Low certainty that increased CT screening contributed to reduced PID at population level: temporal relationship and lag are biologically plausible but design is ecological, with important unmeasured confounding.

4 Aboriginal and Torres Strait Islander populations

4.1 Desirable and Undesirable Effects of Screening

While not directly comparing the impact of different screening intervals, one large stepped-wedge cluster-randomised trial (**Ward et al**)[35] has been conducted among remote Aboriginal communities. The STRIVE trial recruited 68 primary care centres grouped in 24 clusters from 2010-2011, with clusters randomised to transition into the intervention phase during different years of 3 years of follow-up. The study demonstrated that implementing continuous quality-improvement in remote Aboriginal primary health centres was associated with increased STI testing coverage. Across all STIs combined, testing coverage was 38% higher in intervention clusters than controls (22% vs 16%), with greater increases in males (62% increase). This trial indicates that structured, system-level interventions can improve access to screening in remote communities, helping overcome known barriers such as under-testing of men and limited clinic capacity. Despite increases in testing, the intervention did not reduce the prevalence of chlamydia, gonorrhoea, or trichomonas at the community-level after 2 years (RR 0.97; 95% CI 0.84–1.12), indicating no measurable reduction in prevalence from the modest increase in testing.

One modelling study (**Hui et al**)[36] in remote Aboriginal communities suggests that substantially increasing screening coverage could dramatically reduce CT/NG prevalence, especially when combined with highly sensitive point-of-care (POC) tests. Compared to the current assumed background screening of 44% per year, increasing annual screening coverage to 60% was predicted to reduce prevalence to about 3–4% for gonorrhoea and 7% for chlamydia after 10 years, and increasing coverage to 80% could reduce prevalence to around 0.6% and 1.5%, respectively.

4.2 Values, Preferences and Acceptability

Bell et al (2020)[37]

In a qualitative study of 35 young Aboriginal men and women aged 16-21 years, participants' willingness to engage in STI screening was shaped by concerns about privacy, shame, and fear of reputational damage, which made clinic-based testing often emotionally risky and socially unacceptable, especially in communities where being seen at the clinic could trigger gossip or stigma. However, screening was more acceptable when it occurred in culturally safe, discreet, and supportive environments, such as women's groups, outreach activities, or when young people were accompanied by trusted family or peers. Overall, young people valued confidentiality, culturally safe support, and non-judgmental interactions, and acceptability was strongly influenced by social dynamics and health-service design.

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